

NORTH SYDNEY BOYS HIGH SCHOOL

2015 HSC ASSESSMENT TASK 3 (TRIAL HSC)

Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – **3 hours**
- For Section I, shade the correct box on the sheet provided.
- For Section II, write in the booklet provided.
- Each new question is to be started on a **new page**.
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question

- Marks may be deducted for illegible or incomplete working
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Ireland
- ☐ Mr Lin
- ☐ Mr Weiss

Student Number

(To be used by the exam markers only.)

[illegible]

Section I: Objective Response

Mark your answers on the multiple choice answer sheet provided by shading the correct box.

1. One solution to the equation $x^4 - 6x^3 + 26x^2 - 46x + 65 = 0$ is $x = 2 - 3i$.
Another solution is:

(A) $1 - 2i$ (B) $-1 - 2i$ (C) $-2 - i$ (D) $-2 + i$

2. What restrictions must be placed on p if α, β, γ are the three non-zero real roots of the equation $x^3 + px - 1 = 0$?

(A) $p > 0$, p is real (B) $p < 0$, p is real
(C) $p \geq 0$, p is real (D) $p \leq 0$, p is real

3. Consider the two statements:

$$\text{I: } \int_0^1 \frac{dx}{1+x^n} < \int_0^1 \frac{dx}{1+x^{n+1}}$$
$$\text{II: } \int_0^{\frac{\pi}{2}} \sqrt{\sin x} \, dx = \int_0^{\frac{\pi}{2}} \sqrt{\cos x} \, dx$$

Which of following is true?

(A) Neither statement (B) Statement I only
(C) Statement II only (D) Both statements

4. The polynomial equation $x^3 + 4x^2 - 2x - 5 = 0$ has roots α, β, γ .

Which of the following equations has roots $\alpha^2, \beta^2, \gamma^2$?

(A) $x^3 - 20x^2 - 44x - 25 = 0$ (B) $x^3 - 20x^2 + 44x - 25 = 0$
(C) $x^3 - 4x^2 + 5x - 1 = 0$ (D) $x^3 + 4x^2 + 5x - 1 = 0$

5. Which of the following is an expression for $\int \frac{dx}{\sqrt{7-6x-x^2}}$? 1

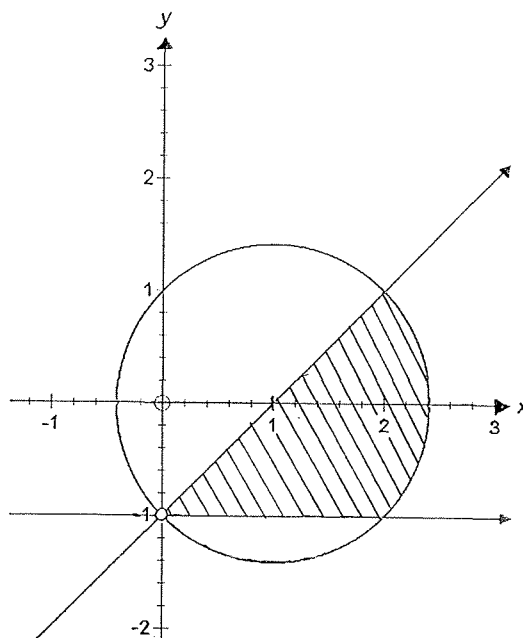
(A) $\sin^{-1}\left(\frac{x-3}{2}\right) + c$

(B) $\sin^{-1}\left(\frac{x+3}{2}\right) + c$

(C) $\sin^{-1}\left(\frac{x-3}{4}\right) + c$

(D) $\sin^{-1}\left(\frac{x+3}{4}\right) + c$

6. The shaded area in the Argand diagram below could be described by which pair of inequalities? 1



(A) $|z-1| \leq \sqrt{2}$ and $0 \leq \arg(z-i) \leq \frac{\pi}{4}$

(B) $|z-1| \leq \sqrt{2}$ and $0 \leq \arg(z+i) \leq \frac{\pi}{4}$

(C) $|z-1| \leq 1$ and $0 \leq \arg(z-i) \leq \frac{\pi}{4}$

(D) $|z-1| \leq 1$ and $0 \leq \arg(z+i) \leq \frac{\pi}{4}$

7. The region bounded by the parabola $y = x^2$ and the x -axis between $x = 0$ and $x = 1$ is rotated about the line $x = 2$ to form a solid of volume V .

Which of the following is an expression for V ?

1

(A) $\pi \int_0^1 (1-x)^2 dy$

(B) $\pi \int_0^1 (1^2 - x^2) dy$

(C) $\pi \int_0^1 [(2-x)^2 - 1^2] dy$

(D) $\pi \int_0^1 [2^2 - (2-x)^2] dy$

8. Which of the following is equal to $\int \sin^3 x \, dx$? 1

(A) $\frac{1}{4} \sin^4 x + c$

(B) $-\cos x + \frac{1}{3} \cos^3 x + c$

(C) $-\cos x - \frac{1}{3} \cos^3 x + c$

(D) $\cos x - \frac{1}{3} \cos^3 x + c$

9. The equation of the tangent to the ellipse $x = 3 \cos \theta$, $y = 2 \sin \theta$ at the point

where $\theta = \frac{\pi}{3}$ is: 1

(A) $6\sqrt{3}x - 4y - 5\sqrt{3} = 0$

(B) $2x - 3\sqrt{3}y - 12 = 0$

(C) $2x + 3\sqrt{3}y - 12 = 0$

(D) $6\sqrt{3}x + 4y - 5\sqrt{3} = 0$

10. $P(4, 25)$ is a point on the rectangular hyperbola $xy = 100$. 1

The tangent at P cuts the hyperbola's asymptotes at Q and R .

The area of $\triangle OQR$ (where O is the origin) is:

(A) $200\sqrt{2} \, u^2$

(B) $2\sqrt{50} \, u^2$

(C) $100 \, u^2$

(D) $200 \, u^2$

Section II: Short Answer

Question 11 (15 marks)

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(a) Evaluate $\int_0^{\frac{\pi}{3}} \sec^4 x \tan x \, dx$ 3

(b) Find $\int \frac{dx}{\sqrt{x^2 - 8x + 25}}$ 2

(c)
(i) Resolve $\frac{9 + x - 2x^2}{(1 - x)(3 + x^2)}$ into partial fractions. 2

(ii) Hence find $\int \frac{9 + x - 2x^2}{(1 - x)(3 + x^2)} \, dx$ 2

(d) Use the substitution $t = \tan\left(\frac{x}{2}\right)$ to evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1}{1 - \cos x} \, dx$ 3

(e)
(i) Prove that $\int_0^a f(x) \, dx = \int_0^a f(a - x) \, dx$ 1

(ii) Hence evaluate $\int_0^{\pi} x \sin x \, dx$ 2

Question 12 (15 marks)

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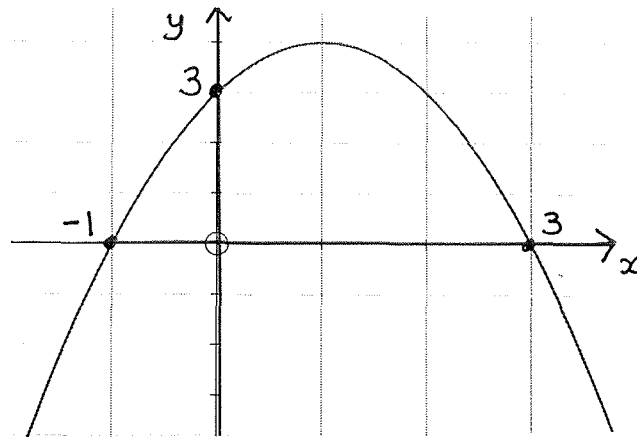
Mark

- (a) Find the square roots of $9 - 40i$. (Give your answer in the form $a + ib$) 2
- (b) Express $z = \sqrt{3} + i$ in modulus-argument form. 1
- (c) (i) Find the Cartesian equation of the locus represented by $2|z| = 3(z + \bar{z})$. 2
- (ii) Sketch the locus on an Argand diagram. 1
- (d) Given that $z = \cos\theta + i \sin\theta$,
- (i) Show that $z^n + z^{-n} = 2\cos n\theta$ 2
- (ii) Hence solve the equation $2z^4 - z^3 + 3z^2 - z + 2 = 0$ 3
- (e) P is a point in the complex plane representing the complex number z , where
- z satisfies $|z - 2| = 2$ and $0 < \arg z < \frac{\pi}{2}$.
- (i) Sketch the locus described by these conditions. 1
- (ii) Find the value of the real number k if $\arg(z - 2) = k \arg(z^2 - 2z)$. 3

Question 13 (15 marks)

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- (a) Let $f(x) = -(x-3)(x+1)$. The graph shown below depicts $y = f(x)$:



On separate diagrams, sketch the following graphs without using calculus.

Indicate any asymptotes, intercepts or other important features.

(i) $y = f(|x|)$ 2

(ii) $y = \frac{1}{f(x)}$ 2

(iii) $y = e^{f(x)}$ 2

(iv) $y^2 = f(x)$ 2

(b)

(i) State the domain and range of $y = \cos^{-1}(e^x)$ 2

(ii) Without using calculus, sketch the graph of $y = \cos^{-1}(e^x)$, showing clearly any intercepts and the equations of any asymptotes. 2

(c) For the curve defined by $3x^2 + y^2 - 2xy - 8x + 2 = 0$ find the coordinates of the points on the curve where the tangent is parallel to the line $y = 2x$. 3

Question 14 (15 marks)

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Mark

(a)

- (i) If α is a root of $P(x)$ with multiplicity n , show that α is also a root of $P'(x)$ with multiplicity $n-1$.

1

- (ii) Given $P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$ has a triple root, factorise $P(x)$ into its linear factors.

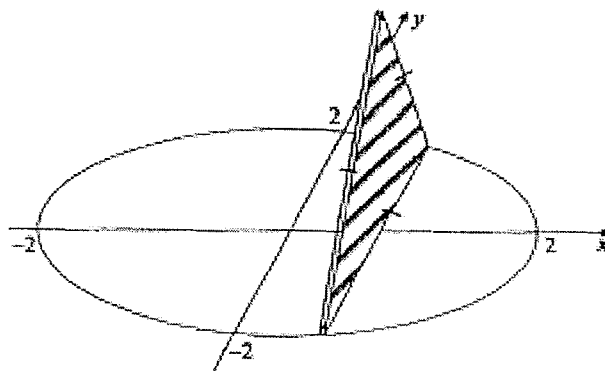
3

- (b) Using the method of cylindrical shells, find the volume generated by revolving the area bounded by the lines $x = \pm 2$ and the hyperbola $\frac{y^2}{9} - \frac{x^2}{4} = 1$ about the y -axis.

4

- (c) The diagram below shows a cross-sectional slice of a solid whose base is the region enclosed by the circle $x^2 + y^2 = 4$. Each such cross-section is an equilateral triangle. Find the volume of the solid.

3



- (d) Suppose that $P(x) = x^4 - 2x^3 + 3x^2 - 4x + 1$ and the equation $P(x) = 0$ has

roots $\alpha, \beta, \gamma, \delta$,

- (i) Show that $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = -2$

2

- (ii) Hence prove that the equation $P(x) = 0$ has precisely two real roots.

2

(a) $P(5p, \frac{5}{p})$ and $Q(5q, \frac{5}{q})$, $p, q > 0$, are two variable points on the hyperbola $xy = 25$.

(i) Derive the equation of the chord PQ . 2

(ii) State the equations of the tangents at P and Q . 1

(iii) If the tangents at P and Q intersect at R , find the coordinates of R . 2

(iv) If the secant PQ passes through the point $(15, 0)$, find the locus of R . 2

(b) Points P and Q are the endpoints of a focal chord of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

If the parameters at P and Q are θ and ϕ respectively, show that the ellipse's

eccentricity is given by $e = \frac{\sin(\theta - \phi)}{\sin\theta - \sin\phi}$. 3

(c) A sequence of numbers T_n , $n = 1, 2, 3, \dots$, is defined by $T_1 = 2$, $T_2 = 0$ and 5

$$T_n = 2T_{n-1} - 2T_{n-2}, \text{ for } n = 3, 4, 5, \dots$$

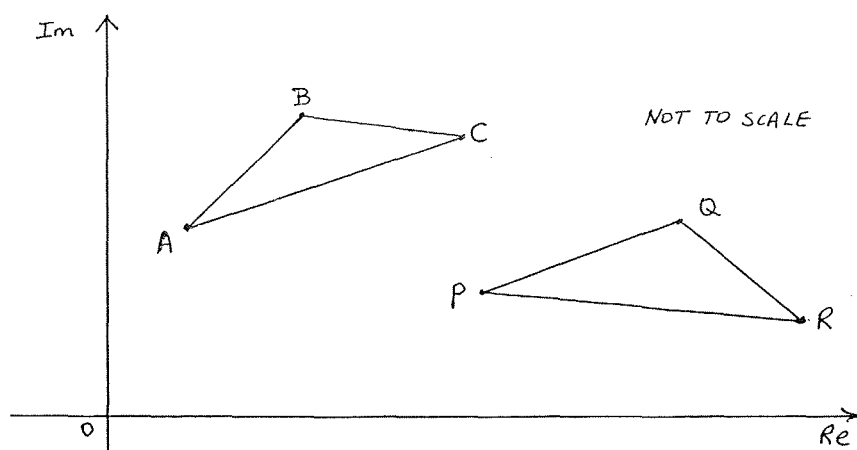
Use mathematical induction to show that $T_n = (\sqrt{2})^{n+2} \cos \frac{n\pi}{4}$, $n = 1, 2, 3, \dots$

Question 16 (15 marks)

Commence a NEW page

Mark

(a)



The points A, B, C , represent the complex numbers z_1, z_2, z_3 respectively.

The points P, Q, R , represent the complex numbers w_1, w_2, w_3 .

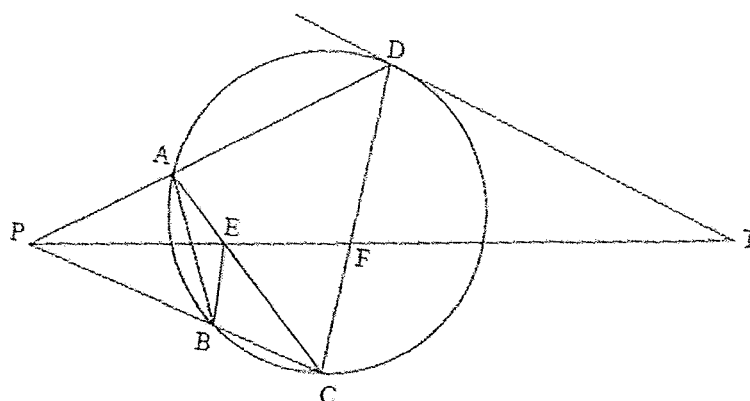
If $\frac{z_2 - z_1}{z_3 - z_1} = \frac{w_2 - w_1}{w_3 - w_1}$ then prove that $\triangle ABC$ is similar to $\triangle PQR$.

3

(b) $ABCD$ is a cyclic quadrilateral. DA produced and CB produced meet at P .

T is a point on the tangent at D to the circle through A, B, C and D .

PT cuts CA and CD at E and F respectively. $TF = TD$.



Copy this diagram into your writing booklet.

(i) Show that $AEFD$ is a cyclic quadrilateral.

2

(ii) Show that $PBEA$ is a cyclic quadrilateral.

3

(c) Let $I_n = \int_0^1 (1 - x^2)^n dx$ and $J_n = \int_0^1 x^2 (1 - x^2)^n dx$

(i) Apply integration by parts to I_n to show that $I_n = 2n J_{n-1}$ 2

(ii) Hence show that $I_n = \frac{2n}{2n+1} I_{n-1}$ 2

(iii) Show that $J_n = I_n - I_{n+1}$ and hence deduce that $J_n = \frac{1}{2n+3} I_n$ 2

(iv) Hence write down a reduction formula for J_n in terms of J_{n-1} 1

END OF EXAMINATION

SECTION 1 - OBJECTIVE RESPONSE

1. A

6. B

2. B

7. C

3. D

8. B

4. B

9. C

5. D

10. D

10

Q.11

$$\begin{aligned}
 \textcircled{a} \quad I &= \int_0^{\pi/3} \sec^4 x \tan x \, dx \\
 &= \int_0^{\pi/3} \sec^3 x \cdot \sec x \tan x \, dx \\
 &= \left[\frac{\sec^4 x}{4} \right]_0^{\pi/3} \quad \text{by standard integrals} \\
 &= \frac{2^4}{4} - \frac{1^4}{4} \\
 &= \frac{15}{4}
 \end{aligned}$$

✓✓ uses
table
correctly✓ correct
answer

$$\begin{aligned}
 \text{ALT: } I &= \int \sec^2 x \cdot \sec^2 x \tan x \, dx \\
 &= \int (1 + \tan^2 x) \cdot \tan x \cdot \sec^2 x \, dx \\
 &= \int (\tan x + \tan^3 x) \sec^2 x \, dx \\
 &= \left[\frac{\tan^2 x}{2} + \frac{\tan^4 x}{4} \right]_0^{\pi/3} \\
 &= \frac{3}{2} + \frac{9}{4} \\
 &= \frac{15}{4}
 \end{aligned}$$

Q11 - ctd

$$(b) \quad I = \int \frac{dx}{\sqrt{x^2 - 8x + 25}}$$

$$= \int \frac{dx}{\sqrt{(x-4)^2 + 3^2}}$$

$$= \ln \left(x-4 + \sqrt{(x-4)^2 + 9} \right) + C$$

(by standard integrals)

✓ completes square

✓ uses table correctly

$$(c) \quad \frac{9+x-2x^2}{(1-x)(3+x^2)} \equiv \frac{A}{1-x} + \frac{Bx+C}{3+x^2}$$

$$(i) \quad \frac{9+x-2x^2}{(1-x)(3+x^2)}$$

$$\begin{aligned} \therefore 9+x-2x^2 &\equiv A(3+x^2) + (Bx+C)(1-x) \\ &= (A-B)x^2 + (B-C)x + (3A+C) \end{aligned}$$

$$\therefore \quad A-B = -2 \quad (1)$$

$$B-C = 1 \quad (2)$$

$$3A+C = 9 \quad (3)$$

$$(1)+(2) \rightarrow A-C = -1 \quad (4)$$

$$(3)+(4) \rightarrow 4A = 8 \quad \therefore A = 2$$

$$\therefore C = 3, \quad B = 4.$$

$$\therefore \frac{9+x-2x^2}{(1-x)(3+x^2)} = \frac{2}{1-x} + \frac{4x+3}{3+x^2}$$

✓✓

$$(ii) \quad I = \int \left[\frac{2}{1-x} + \frac{4x}{3+x^2} + \frac{3}{3+x^2} \right] dx$$

$$\therefore I = -2 \ln|1-x| + 2 \ln(3+x^2) + \sqrt{3} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + C$$

✓✓

Q11 - ctd

$$(d) \quad I = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1}{1 - \cos x} dx \quad \text{Let } t = \tan\left(\frac{x}{2}\right)$$
$$\therefore dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$
$$2dt = (1 + \tan^2 \frac{x}{2}) dx$$
$$\text{ie } dx = \frac{2}{1+t^2} dt$$

$$\begin{cases} x = \frac{\pi}{3} \rightarrow t = \frac{1}{\sqrt{3}} \\ x = \frac{\pi}{2} \rightarrow t = 1 \end{cases}$$

$$\text{Thus } I = \int_{\frac{1}{\sqrt{3}}}^1 \frac{1}{1 - \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$
$$= \int_{\frac{1}{\sqrt{3}}}^1 \frac{1}{t^2} dt$$
$$= \left[-\frac{1}{t} \right]_{\frac{1}{\sqrt{3}}}^1 = -1 + \sqrt{3}.$$

✓ makes
dx substⁿ
& changes
limits

✓ correctly
simplifies

✓ correct
answer

$$(e) (i) \text{ If } I = \int_0^a f(a-x) dx, \quad \text{let } u = a-x$$
$$\therefore du = -dx$$
$$x=0 \rightarrow u=a, \quad x=a \rightarrow u=0$$

$$\therefore I = \int_a^0 f(u) \cdot -du$$
$$= \int_0^a f(u) du = \int_0^a f(x) dx.$$

✓ correct
proof

$$(ii) \quad I = \int_0^{\pi} x \sin x dx = \int_0^{\pi} (\pi-x) \sin(\pi-x) dx \quad \text{from (i)}$$
$$= \int_0^{\pi} \pi \sin x dx - \int_0^{\pi} x \sin x dx$$

✓ uses
part (i)

$$\therefore 2I = \pi \int_0^{\pi} \sin x dx = \pi [-\cos x]_0^{\pi}$$
$$= \pi [1 - (-1)] = 2\pi$$
$$\therefore I = \pi.$$

✓ correct
answer

Q12

(a) Let $z^2 = (a+ib)^2 = 9-40i$

$$\therefore a^2 - b^2 = 9$$

and $2abi = -40i \therefore ab = -20.$

By inspection, $a=5, b=-4$ or $a=-5, b=4$

i.e. square roots are $5-4i$ and $-5+4i$.

(b) $z = \sqrt{3} + i = 2\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$

$$z = 2\left(\cos \frac{\pi}{6} + \sin \frac{\pi}{6} i\right)$$

(c) (i) Let $z = x+iy$, x, y real

$$2|z| = 3(z + \bar{z})$$

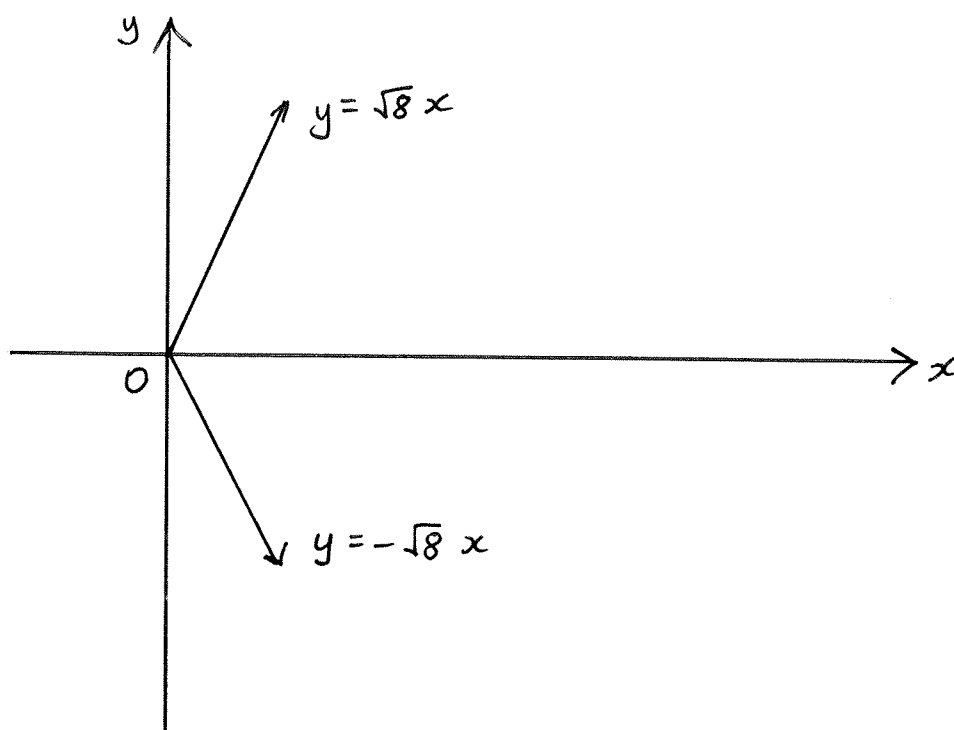
$$4|z|^2 = 9(z + \bar{z})^2 \quad \text{and } x \geq 0$$

$$4(x^2 + y^2) = 9(2x)^2$$

$$4y^2 = 32x^2$$

$$y^2 = 8x^2, \quad x \geq 0$$

(ii)



Q12 - ctd

(d) (i) $z = \cos \theta + i \sin \theta$

$$z^n = (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$z^{-n} = (\cos \theta + i \sin \theta)^{-n} = \cos(-n\theta) + i \sin(-n\theta) \\ = \cos n\theta - i \sin n\theta$$

$$\therefore z^n + z^{-n} = \cancel{\cos n\theta + i \sin n\theta} + \cancel{\cos n\theta - i \sin n\theta} \\ = 2 \cos n\theta$$

✓✓

(ii)

$$2z^4 - z^3 + 3z^2 - z + 2 = 0$$

$$2z^4 + 2 - z^3 - z + 3z^2 = 0$$

$$2(z^4 + 1) - (z^3 + z) + 3z^2 = 0$$

($\div z^2 \rightarrow$) $2(z^2 + z^{-2}) - (z + z^{-1}) + 3 = 0$

$$2(2\cos 2\theta) - 2\cos \theta + 3 = 0$$

$$4\cos 2\theta - 2\cos \theta + 3 = 0$$

$$4(2\cos^2 \theta - 1) - 2\cos \theta + 3 = 0$$

$$8\cos^2 \theta - 2\cos \theta - 1 = 0$$

$$(2\cos \theta - 1)(4\cos \theta + 1) = 0$$

$$\cos \theta = \frac{1}{2} \quad \text{or} \quad \cos \theta = -\frac{1}{4}$$

if $\cos \theta = \frac{1}{2}$, $\sin \theta = \pm \frac{\sqrt{3}}{2}$

if $\cos \theta = -\frac{1}{4}$, $\sin \theta = \pm \frac{\sqrt{15}}{4}$

$\therefore$ roots are $z = \frac{1}{2} \pm i \frac{\sqrt{3}}{2}$, $-\frac{1}{4} \pm i \frac{\sqrt{15}}{4}$.

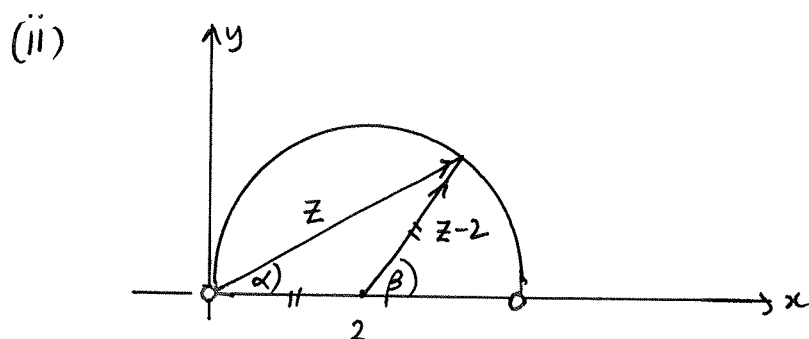
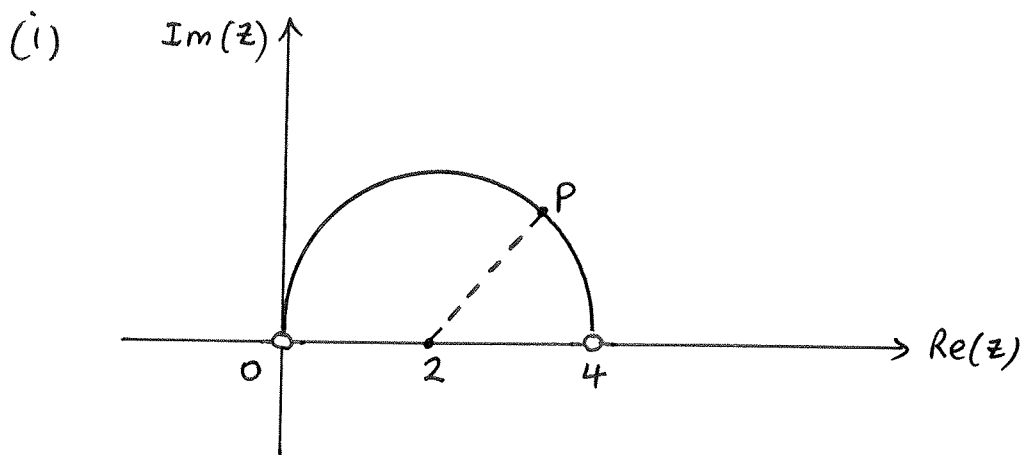
✓

✓

✓

Q12) ctd

(e) $|z-2| = 2$ and $0 < \arg z < \frac{\pi}{2}$



$$\begin{aligned} \arg(z-2) &= k \arg(z^2 - 2z) \\ &= k \arg z(z-2) \\ &= k(\arg z + \arg(z-2)) \end{aligned}$$

using vectors, we see that

$$\arg(z-2) = 2 \cdot \arg z \quad \left(\begin{array}{l} \text{exterior } \angle \text{ of } \Delta; \\ \text{equal radii} \end{array} \right).$$

Hence $2 \arg z = k(\arg z + 2 \arg z)$

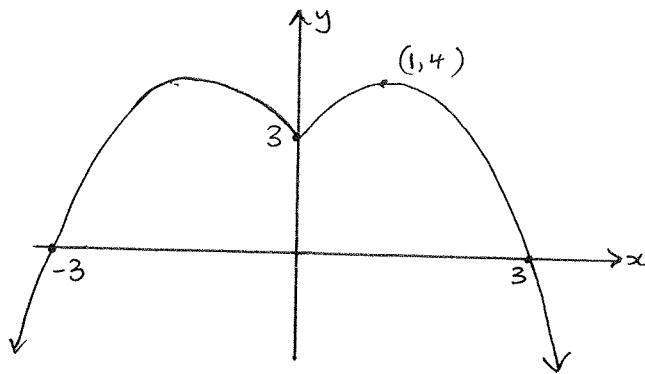
$$2 = 3k$$

$$\therefore k = \frac{2}{3}$$

Q13

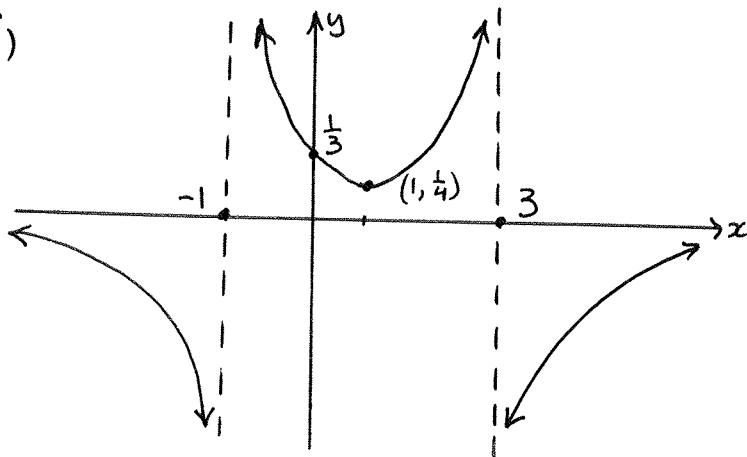
$$f(x) = -(x-3)(x+1)$$

(a)
(i)



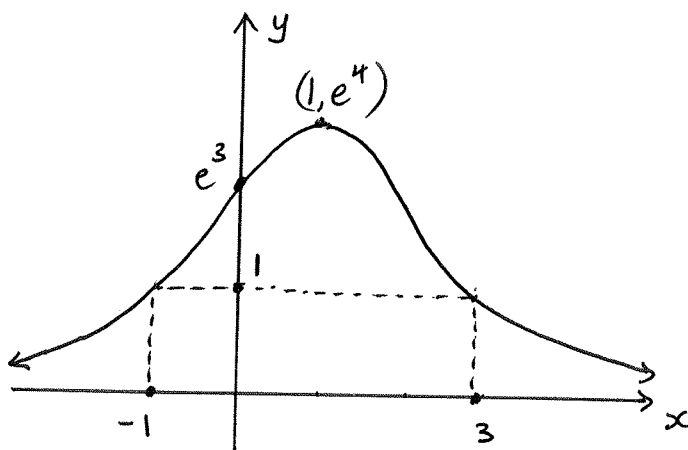
✓✓

(ii)



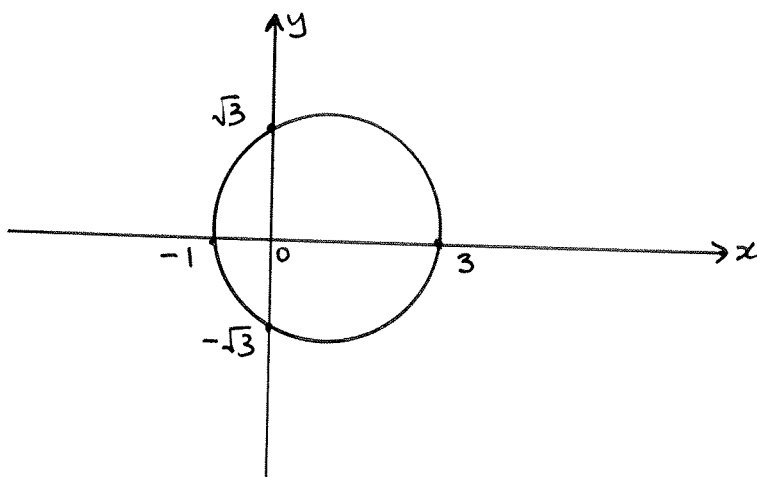
✓✓

(iii)



✓✓

(iv)



✓✓

Q13) ctd.

(b) (i) $y = \cos^{-1}(e^x)$

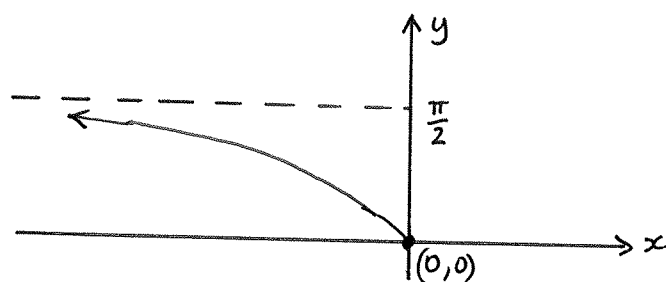
We need $-1 \leq e^x \leq 1 \quad \therefore e^x \leq 1 \quad \therefore x \leq 0$

$\therefore D: x \leq 0$

Since $x \leq 0$, $0 \leq y < \frac{\pi}{2}$ (as $e^x \rightarrow 0^+$ as $x \rightarrow -\infty$)

$\therefore R: 0 \leq y < \frac{\pi}{2}$

(ii)



$y = \cos^{-1}(e^x)$.
($y = \frac{\pi}{2}$ is asymptote).

(c) $3x^2 + y^2 - 2xy - 8x + 2 = 0$

$\therefore 6x + 2y \frac{dy}{dx} - 2y - 2x \frac{dy}{dx} - 8 = 0$

$\therefore \frac{dy}{dx} = \frac{8 - 6x + 2y}{2y - 2x} = \frac{4 - 3x + y}{y - x}$

Thus $\frac{4 - 3x + y}{y - x} = 2$

$\therefore 4 - 3x + y = 2y - 2x \quad \therefore y = 4 - x$

Sub. in curve equation:-

$3x^2 + 16 - 8x + x^2 - 8x + 2x^2 - 8x + 2 = 0$

$6x^2 - 24x + 18 = 0$

$x^2 - 4x + 3 = 0$

$(x-3)(x-1) = 0$

$\therefore x=3 \rightarrow y=1$

or $x=1 \rightarrow y=3$

$\therefore$ the points are $(3,1)$ and $(1,3)$.

Q14

(a) (i) Let $P(x) = (x-\alpha)^n \cdot Q(x)$, $Q(\alpha) \neq 0$.

$$\begin{aligned}\therefore P'(x) &= n(x-\alpha)^{n-1} \cdot Q(x) + (x-\alpha)^n \cdot Q'(x) \\ &= (x-\alpha)^{n-1} [n \cdot Q(x) + (x-\alpha) \cdot Q'(x)] \\ &= (x-\alpha)^{n-1} \cdot Q_1(x), \text{ where } Q_1(\alpha) \neq 0\end{aligned}$$

$\therefore \alpha$ is a root of $P'(x)$ of multiplicity $n-1$. ✓

(ii) $P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$ has a triple root

$\therefore P'(x) = 8x^3 + 27x^2 + 12x - 20$ has a double root

$\therefore P''(x) = 24x^2 + 54x + 12$ has a 1-fold root. ✓

$$\therefore 24x^2 + 54x + 12 = 0$$

$$4x^2 + 9x + 2 = 0$$

$$(4x+1)(x+2) = 0$$

$$\therefore x = -\frac{1}{4} \text{ or } -2. \quad \checkmark$$

Subbing $x = -2$,

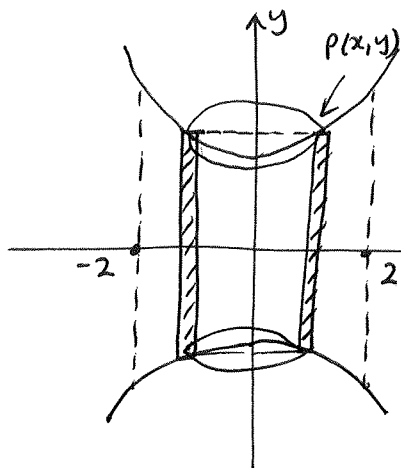
$$P(-2) = 2(-2)^4 + 9(-2)^3 + 6(-2)^2 - 20(-2) - 24 = 0$$

$\therefore x = -2$ is the triple root.

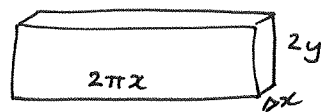
By inspection, $P(x) = (x+2)^3 \cdot (2x-3)$ ✓

Q14 - ctd

(b)



A shell at $P(x, y)$ has height $2y$ and curved surface area $2\pi x$.



$$\text{Thus } \Delta V \doteq 2\pi x \cdot 2y \cdot \Delta x$$

$$\text{Now } \frac{y^2}{9} - \frac{x^2}{4} = 1 \quad \therefore y^2 = 9\left(1 + \frac{x^2}{4}\right)$$

$$\therefore y = \pm \frac{3}{2} \sqrt{4+x^2}$$

✓

$$\therefore \Delta V = 2\pi x \cdot 3\sqrt{4+x^2} \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{x=2} 6\pi x \sqrt{4+x^2} \Delta x$$

$$V = 6\pi \int_0^2 x \sqrt{4+x^2} dx$$

✓

$$= 3\pi \int_0^2 2x \sqrt{4+x^2} dx$$

$$= 3\pi \left[\frac{2}{3} (4+x^2)^{\frac{3}{2}} \right]_0^2$$

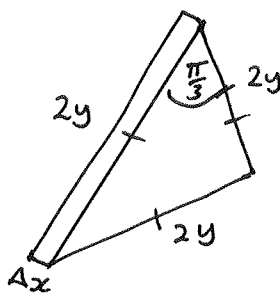
✓

$$V = 2\pi \left[8^{\frac{3}{2}} - 4^{\frac{3}{2}} \right]$$

$$\therefore V = 16\pi (2\sqrt{2} - 1) u^3$$

✓

Q14 - ctd. (C)



Area of each cross-sectional slice is $\frac{1}{2} (2y)^2 \sin \frac{\pi}{3} = \sqrt{3} y^2$

$$\therefore \Delta V \doteq \sqrt{3} y^2 \cdot \Delta x$$

$$\therefore \Delta V \doteq \sqrt{3} (4-x^2) \Delta x$$

$$\text{Thus } V = \lim_{\Delta x \rightarrow 0} \sum_{x=-2}^{+2} \sqrt{3} (4-x^2) \Delta x$$

$$V = \sqrt{3} \int_{-2}^2 (4-x^2) dx = 2\sqrt{3} \int_0^2 (4-x^2) dx$$

$$\therefore V = 2\sqrt{3} \left[4x - \frac{x^3}{3} \right]_0^2$$

$$= 2\sqrt{3} \left(8 - \frac{8}{3} \right)$$

$$\therefore V = \frac{32\sqrt{3}}{3} u^3.$$

(d) $P(x) = x^4 - 2x^3 + 3x^2 - 4x + 1$

(i) $\alpha + \beta + \gamma + \delta = -(-2) = 2$

$(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) = 3$

So $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$

$$= 2^2 - 2(3)$$

$$= -2$$

(ii) Since $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 < 0$

$\therefore$ at least one of $\alpha, \beta, \gamma, \delta$ is non-real.

But since the coefficients of $P(x)$ are real, the conjugate of this non-real root is also a root. What of the remaining 2 roots?

We observe that $P(-1) = 1 + 2 + 3 + 4 + 1 = 11 > 0$

and $P(1) = 1 - 2 + 3 - 4 + 1 = -1 < 0$.

$\therefore$ Since $P(x)$ is continuous, and $P(-1)$ and $P(1)$ have opposite signs, $y = P(x)$ crosses the x axis $\therefore P(x)$ has a real root. But the remaining root must also be real, else it would have a conjugate that's a root. $\therefore$ exactly 2 real roots.

Q15 (a) $P(5p, \frac{5}{p})$, $Q(5q, \frac{5}{q})$; $p, q > 0$

$$(i) m_{PQ} = \frac{\frac{5}{q} - \frac{5}{p}}{5q - 5p} = -\frac{1}{pq}$$

$$\therefore PQ \text{ equation is } y - \frac{5}{p} = -\frac{1}{pq}(x - 5p)$$

$$\therefore pqy - 5q = -x + 5p$$

$$\therefore x + pqy = 5(p+q)$$

(ii) For tangent at P, let $q \rightarrow p$

$$\therefore x + p^2y = 5(2p)$$

$$\therefore x + p^2y = 10p$$

Likewise, tangent at Q is $x + q^2y = 10q$.

(iii) For R, solve tangents simultaneously:

$$y(p^2 - q^2) = 10(p - q)$$

$$\therefore \text{as } p \neq q, \quad y = \frac{10}{p+q}$$

$$\text{Thus } x = 10p - \frac{10p^2}{p+q} = \frac{10pq}{p+q}$$

$$\therefore R = \left(\frac{10pq}{p+q}, \frac{10}{p+q} \right)$$

$$(iv) PQ \text{ secant is } x + pqy = 5(p+q)$$

$$\therefore \text{if goes thru } (15, 0), \quad 15 + 0 = 5(p+q)$$

$$\therefore p+q = 3$$

$$\text{Thus } R = \left(\frac{10pq}{3}, \frac{10}{3} \right)$$

$$\text{So locus is } y = \frac{10}{3}$$

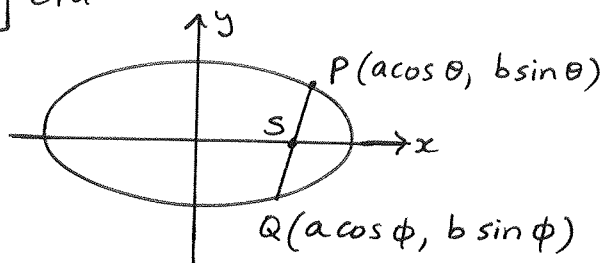
But $x > 0$, since $p, q > 0$, and since the intersection cannot occur inside the hyperbola, $\therefore x < 7.5$.

$$\therefore \text{Locus is } y = \frac{10}{3}, \quad 0 < x < 7.5$$

[Note: the tangent at $(7.5, \frac{10}{3})$ on H goes thru $(15, 0)$]

Q15 - ctd.

(b)



$$S = (ae, 0).$$

Since PQ is a focal chord, S lies on PQ.

$$\therefore m_{PS} = m_{PQ}$$

$$\therefore \frac{b \sin \theta - 0}{a \cos \theta - ae} = \frac{b \sin \theta - b \sin \phi}{a \cos \theta - a \cos \phi}$$

$$\therefore \frac{\cancel{b} \sin \theta}{\cancel{a} (\cos \theta - e)} = \frac{\cancel{b} (\sin \theta - \sin \phi)}{\cancel{a} (\cos \theta - \cos \phi)}$$

$$\begin{aligned} \therefore \cancel{\sin \theta \cos \theta} - \sin \theta \cos \phi \\ = \cancel{\cos \theta \sin \theta} - \cos \theta \sin \phi - e (\sin \theta - \sin \phi) \end{aligned}$$

$$\therefore -\sin \theta \cos \phi + \cos \theta \sin \phi = -e (\sin \theta - \sin \phi)$$

$$\therefore \sin \theta \cos \phi - \cos \theta \sin \phi = e (\sin \theta - \sin \phi)$$

$$\therefore \sin(\theta - \phi) = e (\sin \theta - \sin \phi)$$

$$\therefore e = \frac{\sin(\theta - \phi)}{\sin \theta - \sin \phi}, \text{ as required.}$$

[ALT. Equation of PQ is

$$y - b \sin \theta = \frac{b \sin \theta - b \sin \phi}{a \cos \theta - a \cos \phi} (x - a \cos \theta).$$

Sub. in $S(ae, 0)$, and rearrange correctly.]

Q15 - ctd.

(c) We have $T_n = 2T_{n-1} - 2T_{n-2}$, $n = 3, 4, 5 \dots$

and $T_1 = 2$, $T_2 = 0$. To prove: $T_n = (\sqrt{2})^{n+2} \cos \frac{n\pi}{4}$.

When $n=1$: $(\sqrt{2})^{1+2} \cos \frac{1(\pi)}{4} = 2\sqrt{2} \cdot \frac{1}{\sqrt{2}} = 2$

$\therefore$ true for $n=1$

When $n=2$: $(\sqrt{2})^{2+2} \cos \frac{2\pi}{4} = 4 \times 0 = 0$

$\therefore$ true for $n=2$

✓ proves for $n=1, 2$

Assume true for $n \leq k$

i.e. assume $T_n = (\sqrt{2})^{n+2} \cos \frac{n\pi}{4}$, $n = 1, 2, 3, \dots, k$.

Then look at $n = k+1$:

$$T_{k+1} = 2 \cdot T_k - 2 \cdot T_{k-1} \quad \text{by the recursive definition.}$$

$$= 2(\sqrt{2})^{k+2} \cos \frac{k\pi}{4} - 2(\sqrt{2})^{(k-1)+2} \cos \frac{(k-1)\pi}{4}$$

by our assumption.

$$= (\sqrt{2})^{k+3} \left[\sqrt{2} \cos \frac{k\pi}{4} - \cos \left(\frac{k\pi}{4} - \frac{\pi}{4} \right) \right]$$

$$= (\sqrt{2})^{k+3} \left[\frac{2}{\sqrt{2}} \cos \frac{k\pi}{4} - \left(\cos \frac{k\pi}{4} \cos \frac{\pi}{4} + \sin \frac{k\pi}{4} \sin \frac{\pi}{4} \right) \right]$$

$$= (\sqrt{2})^{k+3} \left[\frac{2}{\sqrt{2}} \cos \frac{k\pi}{4} - \frac{1}{\sqrt{2}} \cos \frac{k\pi}{4} - \frac{1}{\sqrt{2}} \sin \frac{k\pi}{4} \right]$$

$$= (\sqrt{2})^{k+3} \left[\frac{1}{\sqrt{2}} \cos \frac{k\pi}{4} - \frac{1}{\sqrt{2}} \sin \frac{k\pi}{4} \right]$$

$$= (\sqrt{2})^{k+3} \left[\cos \frac{\pi}{4} \cos \frac{k\pi}{4} - \sin \frac{\pi}{4} \sin \frac{k\pi}{4} \right]$$

$$= (\sqrt{2})^{k+3} \left[\cos \left(\frac{k\pi}{4} + \frac{\pi}{4} \right) \right]$$

$$\therefore T_{k+1} = (\sqrt{2})^{(k+1)+2} \cos \frac{(k+1)\pi}{4}$$

$\therefore$ if true for $n = 1, 2, \dots, k$ it's true for $n = k+1$.

Hence, by induction & (*), true for $n = 1, 2, 3, \dots$

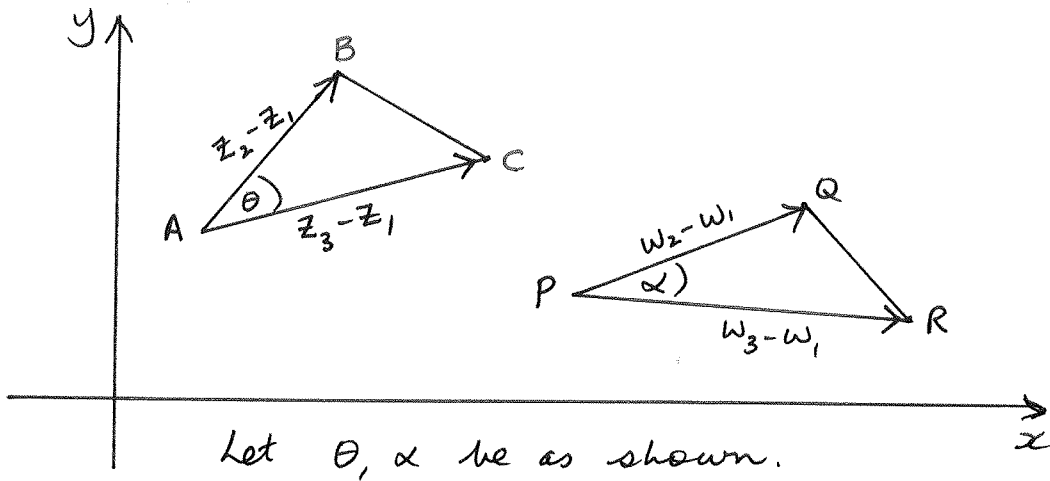
✓ uses definition

✓ rewrites in cosines

✓✓

Q16

(a)



$$\text{Given } \frac{z_2 - z_1}{z_3 - z_1} = \frac{w_2 - w_1}{w_3 - w_1}$$

$$\therefore \left| \frac{z_2 - z_1}{z_3 - z_1} \right| = \left| \frac{w_2 - w_1}{w_3 - w_1} \right|$$

$$\therefore \frac{|z_2 - z_1|}{|z_3 - z_1|} = \frac{|w_2 - w_1|}{|w_3 - w_1|}$$

$$\therefore \frac{AB}{AC} = \frac{PQ}{PR}$$

✓ uses modulus

$$\text{Also, } \text{Arg} \left(\frac{z_2 - z_1}{z_3 - z_1} \right) = \text{Arg} \left(\frac{w_2 - w_1}{w_3 - w_1} \right)$$

$$\therefore \text{Arg} (z_2 - z_1) - \text{Arg} (z_3 - z_1) = \text{Arg} (w_2 - w_1) - \text{Arg} (w_3 - w_1)$$

$$\therefore \theta = \alpha$$

✓ uses argument

Thus $\triangle ABC \parallel \triangle PQR$, since

2 pairs of corresponding sides are in the same ratio, and the included angle is equal.

✓ correct geometrical reason

$$T_D = T_F \quad (\text{given})$$

$\angle TDF = \angle CAD$ (angle between tangent and chord at point of contact equals angle in alternate segment)

$\therefore$ AEFD is cyclic (exterior angle equals interior opposite angle) #

$\angle PBA = \angle ADF$ (exterior angle of cyclic quad ABCD equals interior opposite angle)

$\therefore$ PBEA is cyclic (interval AP subtends equal angles at points E & B which are on the same side of AP). #

Q16] ctd

$$(c) \quad I_n = \int_0^1 (1-x^2)^n dx \quad \& \quad J_n = \int_0^1 x^2 (1-x^2)^n dx$$

$$(i) \quad I_n = \int_0^1 (1-x^2)^n \cdot \frac{d}{dx}(x) dx$$

$$= \left[x (1-x^2)^n \right]_0^1 - \int_0^1 -2nx^2 (1-x^2)^{n-1} dx$$

$$= 0 + 2n \int_0^1 x^2 (1-x^2)^{n-1} dx$$

$$\therefore I_n = 2n \cdot J_{n-1}$$

(ii) from (i),

$$I_n = 2n \int_0^1 x^2 (1-x^2)^{n-1} dx$$

$$= -2n \int_0^1 [(1-x^2)-1] \cdot (1-x^2)^{n-1} dx$$

$$= -2n \cdot (I_n - I_{n-1})$$

$$= -2n \cdot I_n + 2n \cdot I_{n-1}$$

$$\therefore (2n+1) \cdot I_n = 2n \cdot I_{n-1}$$

$$\therefore I_n = \frac{2n}{2n+1} \cdot I_{n-1}$$

Q16- ctd.

(c) (iii)

$$J_n = \int_0^1 x^2 (1-x^2)^n dx$$

$$= - \int_0^1 [(1-x^2)-1] (1-x^2)^n dx$$

$$= - (I_{n+1} - I_n)$$

$$\therefore J_n = I_n - I_{n+1}$$

$$= I_n - \frac{2(n+1)}{2(n+1)+1} \cdot I_n \quad \text{from (ii)}$$

$$= \frac{(2n+3) \cdot I_n - (2n+2) I_n}{2n+3}$$

$$\therefore J_n = \frac{1}{2n+3} \cdot I_n$$

(iv) using (iii) and (i),

$$J_n = \frac{1}{2n+3} \cdot I_n$$

$$= \frac{1}{2n+3} \cdot 2n \cdot J_{n-1}$$

$$\text{i.e. } J_n = \frac{2n}{2n+3} \cdot J_{n-1}$$